

Difference between Homoscedasticity and Heteroscedasticity

Homoscedasticity refers to constant variance in regression errors (residuals) across all independent variable levels, indicating reliable, evenly spread data. Heteroscedasticity occurs when this variance is inconsistent, causing non-uniform error distribution (e.g., a "cone" shape). Homoscedasticity is required for stable linear regression; heteroscedasticity yields inefficient, unreliable estimates.

Key Differences:

- **Definition:** [Homoscedasticity](#) means equal variance, whereas [heteroscedasticity](#) means unequal variance.
- **Regression Assumption:** Homoscedasticity is an assumption of Ordinary Least Squares (OLS) regression.
- **Data Scatter:** Homoscedastic data shows a consistent, uniform spread of data points around the regression line. Heteroscedastic data often displays a "fan" or "cone" shape where variance increases or decreases as the independent variable changes.
- **Consequences:** Heteroscedasticity leads to biased standard errors, unreliable

Visual Comparison:

- **Homoscedasticity Plot:** Data points form a cylinder-like shape, spreading evenly along the fitted regression line.
- **Heteroscedasticity Plot:** Data points spread out (or converge) in a cone shape, indicating higher variance at certain levels.

When They Matter:

- **Homoscedasticity** is ideal for linear models, ensuring that the model's predictive power is similar across the entire data range.
- **Heteroscedasticity** often occurs in financial or behavioural data, such as mapping income vs. luxury spending (higher incomes exhibit more spending variety).

Addressing Heteroscedasticity:

If detected (e.g., using a Breusch-Pagan test or checking residual plots), common solutions include transforming the dependent variable (e.g., log transformation) or using weighted least squares regression